MAE 3240

Spring 2026

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Pset 4 3/3/2026

Acknowledgement: I refered to Course notes 7/8 and to charts in "Fundamentals of Heat and Mass Transfer"

Problem 1

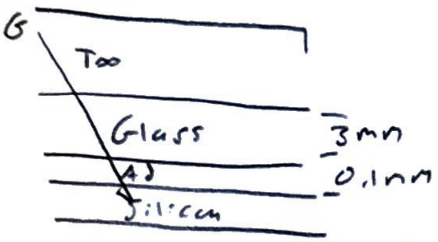
Known: 3mm Glass 0.1mm Ad $h = 10\,W/m^2K$ $h_{rad} = 5\,W/m^2K$
$k_g = 1.5\,W/mK$ $k_a = 1\,W/mK$ $a = 0.553$ $b = 0.001$

$R_c = 10^{-4}\,m^2K/W$ $G = 800\,W/m^2$ $T_\infty = 300 = T_{sur}$

Find:

Thermal Resistance Network, T_{gTop}, T_{si}, η, P_{pwr}

Schematic

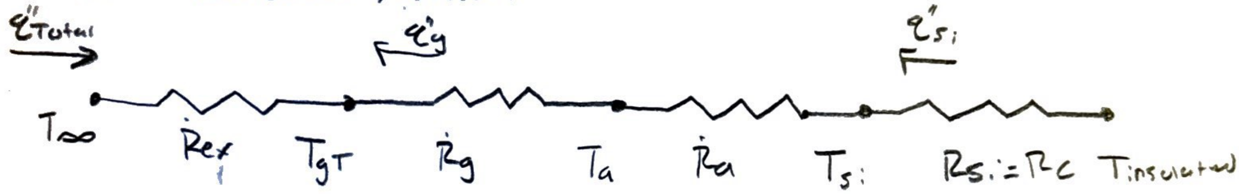


Assumptions

Steady state conditions, 1D convection
No radiation heat loss

Analysis:

Thermal Resistance Network



$$R_{ex} = \frac{1}{h + h_{rad}} = \frac{1}{15}\,m^2K/W \qquad R_g = \frac{0.003}{1.5} = 0.002\,m^2K/W$$

$$R_a = \frac{1\times10^{-4}}{1} = 1\times10^{-4}\,m^2K/W \qquad R_c = 10^{-4} \qquad R_{Total} = 0.002 + 1\times10^{-4} + 1\times10^{-4} + \frac{1}{15} = 0.0689$$

$$q''_g = 0.1\times800 = 80\,J \qquad q''_{si} = (0.85\cdot800) - (0.85\cdot800(a - bT_{si}))$$

$$q''_T = \frac{T_{si} - T_\infty}{R_{Total}} \qquad \Rightarrow 680 - 680(0.553 - 0.001T_{si})$$

$$\Rightarrow 303.96 - 0.68T_{si}$$

$$q''_T = q''_g + q''_{si} \Rightarrow \frac{T_{si} - 300}{0.0689} = 303.96 - 0.68T_{si} + 80 \Rightarrow 26.45 + 0.0469T_{si} = T_{si} - 300$$

$$= 326.45 = 0.9531\,T_{si} \qquad T_{si} = 342.5\,K$$

$$q''_T = \frac{T_{gT} - T_\infty}{R_{ex}} \Rightarrow \frac{342.5 - 300}{0.0689} = \frac{T_{gT} - 300}{1/15} \Rightarrow T_{gT} = 341.12\,K$$

c) $\eta = a - bT_{si} = 0.553 - 0.001(342.5) = 0.2105$

$P_{flux} = 0.2105 \cdot 680 = 143\ W/m^2$

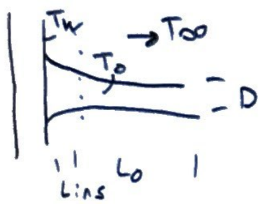
Problem 2

Known $D = 10mm$ $K = 80\ W/mK$ $T_w = 200°C$ $L_{ins} = 200mm$ $L_{exp} = L_o$

$T_{max} = 100°C$ $T_\infty = 20°C$ $h = 20\ W/m^2K$

Find: T_o, Min L_o

Schematic



Assumptions

Steady-State Conditions Insulated end at rod

Constant properties

Negligible Radiation

Analysis:

$-k\frac{\partial T}{\partial x}\Big|_{x=L} = 0$ $P = \pi D = 0.0314$ $A_c = \pi D^2/4 = 7.85\times10^{-5}$ $\theta(x) = T(x) - T_\infty$ $m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{20 \cdot 0.0314}{80 \cdot 7.85\times10^{-5}}} = 10$

$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL}$ $\theta_b = T_b - T_\infty = 200°C - 20°C = 180°C$

$q_f = M\tanh mL = \sqrt{hPkA_c}\,\theta_b \cdot \tanh(mL_o)$ $q_{cond} = \frac{-kA_c}{L_{in}}(T_w - T_o)$

$q_{cond} = q_f \Rightarrow \frac{-kA_c}{L_{in}}(T_w - T_o) = \sqrt{hPkA_c}\,(T_o - T_\infty) \cdot \tanh(mL_o)$

$T_o = \left(\frac{\sqrt{hPkA_c}\,(T_o - T_\infty)\tanh(mL_o)L_{in}}{-kA_c} - T_w\right)$

b) $L_o = 200mm$

$T_o = \frac{-\sqrt{20 \cdot 0.0314 \cdot 80 \cdot 7.85\times10^{-5}}\,(T_o - T_\infty) \cdot \tanh(10 \cdot 0.2)(0.2)}{-80 \cdot 7.85\times10^{-5}} + 200°C$

$\Rightarrow \frac{-0.0628(T_o - T_\infty) \cdot 0.964 \cdot 0.2}{-80 \cdot 7.85\times10^{-5}} + 200°C \Rightarrow -1.928\,T_o + 1.928(20°) + 200$

$T_o = 13.164 + 68.3 \Rightarrow 81.47°C$ Yes it meets requirement c)

$$100 = -\frac{\sqrt{20\cdot 0.0314\cdot 80\cdot 7.85\times10^{-5}\ (100-20)}\cdot \tanh(10\cdot L_0)\cdot 0.2}{-80\times 7.85\times10^{-5}} + 200°C$$

$$0.628 = 0.0628(80)\tanh(10\cdot L_0)\cdot 0.2$$

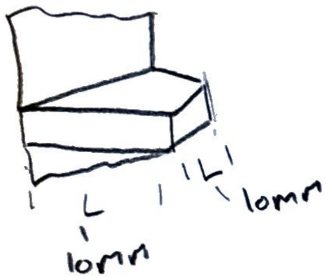
$$-0.625 = \tanh(10\cdot L_0) \Rightarrow 0.733 = 10\cdot L_0 \Rightarrow L_0 = 73mm$$

Problem 3

Known: $k = 400\ W/mK$ $S = 10mm$ $h = 100\ W/m^2K$ $\eta_f = 0.6$

Find: Fin length L, R_f, ε_f

Schematic



Assumptions

Steady-State Conditions No radiation to surrounding

1-D heat transfer

Temp Uniform

Analysis

Course Notes Pin fin

$$\eta_f = \frac{\tanh mL_c}{mL_c} \qquad L = L_c - D/4$$

$$0.6 = \frac{\tanh\left(\sqrt{\frac{100\cdot(2\cdot0.01+2\cdot0.01)}{400\cdot 0.01^2}}\cdot L_c\right)}{\sqrt{\frac{100\cdot(2\cdot0.01+2\cdot0.01)}{400\cdot0.01^2}}\cdot L_c} \Rightarrow 0.6 = \frac{\tanh(10L_c)}{L_c} \Rightarrow$$

$$0.6 = \frac{\tanh(x)}{x} \Rightarrow x \approx 1.51 \quad L_c = 0.151m$$

$$L = 0.151m - \frac{0.01}{4} = 0.1485m = 148.5mm$$

$$\varepsilon_f = \frac{\eta_f h A_s \theta_b}{h A_c \theta_b} \Rightarrow \eta_f \frac{A_s}{A_c}$$

$$= \frac{0.6\cdot\pi\cdot0.01\cdot0.151}{0.01^2}$$

$$= 28.46$$

b)

$$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{0.6\cdot100\cdot\pi\cdot0.01\cdot0.151} \Rightarrow 3.51\ K/W$$

Assignment 4 Reflection:

From this assignment I learned how to use thermal resistance in order to calculate different temperatures between surfaces. The use of thermal resistance, specifically in this instance solar panels, is relevant to modern-day research into renewable energy and its uses. Also, this assignment covered heat dissipated through fins and the temperature associated with convective fins. Cooling and convective fins are common among engineering equipment as a means of protection from overheating and as an effective means to cool equipment. By learning how to use length in order to meet requirements regarding temperature, I have learned more about a system utilized in factories, reactors, and power plants. This is a core process for many systems that are prone to overheating and therefor will be extremely useful in more advanced system design.